MATH 526 Section 001 Lab 7 Spring 2006

Part I

We discuss in this part an algorithm that is based on the **Sherman-Morrison** formula.

Let $\mathbf{A} = \mathbf{B} - \mathbf{u}\mathbf{v}^{\mathrm{t}}$, where $\mathbf{B}$ is a nonsingular $n \times n$ matrix and $\mathbf{u}$, $\mathbf{v}$ are column n-vectors. Then $\mathbf{A}$ is nonsingular if and only if $1 - \mathbf{v}^{\mathrm{t}}(\mathbf{B}^{-1}\mathbf{u}) \neq 0$, in which case the inverse of $\mathbf{A}$ is given by the Sherman-Morrison formula:

$$\mathbf{A}^{-1} = \left[\mathbf{I} + \left(\frac{1}{1 - \mathbf{v}^{\mathrm{t}}(\mathbf{B}^{-1}\mathbf{u})}\right)(\mathbf{B}^{-1}\mathbf{u})\mathbf{v}^{\mathrm{t}}\right]\mathbf{B}^{-1}$$

Based on this formula we have the following algorithm for solving $\mathbf{A}\mathbf{x} = \mathbf{b}$:

1. Solve $\mathbf{B}\mathbf{z} = \mathbf{u}$ and compute $\gamma = 1 - \mathbf{v}^{\mathrm{t}}\mathbf{z}\,\big(= 1 - \mathbf{v}^{\mathrm{t}}(\mathbf{B}^{-1}\mathbf{u})\big)$. If $\gamma = 0$, then $\mathbf{A}$ is singular. If $\gamma \neq 0$, then $\mathbf{A}$ is nonsingular and we proceed to step 2.
2. Solve $\mathbf{B}\mathbf{y} = \mathbf{b}$ and compute $\beta = (\mathbf{v}^{\mathrm{t}}\mathbf{y})/\gamma$.
3. $\mathbf{x} = \mathbf{y} + \beta\mathbf{z}$ is then the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Suppose we have a good solver for $\mathbf{B}$. Then this algorithm will solve $\mathbf{A}\mathbf{x} = \mathbf{b}$ by using the good solver for $\mathbf{B}$, with the same order of efficiency.

Consider the system

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 2 \\ 2 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}.$$

Let $\mathbf{A}$ be the matrix on the left-hand side, and $\mathbf{b}$ be the vector on the right-hand side. Then $\mathbf{A} = \mathbf{B} - \mathbf{u}\mathbf{v}^{\mathrm{t}}$ where

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Create $\mathbf{A}$, $\mathbf{B}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{b}$.

>>A–(B–u∗v′)

This is to check that we have the correct $\mathbf{B}$, $\mathbf{u}$ and $\mathbf{v}$.

>>z=forward(B,u)

forward.m is the M-file that solves a lower triangular system by forward substitution.

>>gamma=1–v′∗z

Since $\gamma \neq 0$, $\mathbf{A}$ is nonsingular and we proceed to solve $\mathbf{A}\mathbf{x} = \mathbf{b}$.

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>>y=forward(B,b)
>>beta=(v'*y)/gamma
>>x=y+beta*z
>>A*x-b
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This is to check that we have solved $\mathbf{Ax} = \mathbf{b}$.

Clear all variables before you work on Part II.